

Medha Agarwal

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EDUCATION

DEC 2026	PhD, Statistics UNIVERSITY OF WASHINGTON ADVISOR: Alex Luedtke
MAY 2021	Bachelor of Science, Mathematics and Scientific Computing INDIAN INSTITUTE OF TECHNOLOGY KANPUR ADVISOR: Dootika Vats MINOR: English Literature

RESEARCH INTERESTS

Causal inference and semiparametric efficiency theory; optimal transport and distributional inference; uncertainty quantification and conformal prediction, particularly under distribution shift; gradient flows and mean-field dynamics of learning algorithms; machine learning for scientific applications, particularly ecology.

PEER-REVIEWED PUBLICATIONS

- [1] **Medha Agarwal** and A. Luedtke. Sinkhorn treatment effects: A causal optimal transport measure. In *Forty-third International Conference on Machine Learning (ICML)*, 2026
- [2] **Medha Agarwal**, K. Rafiq, R. Mehta, B. Abrahms, and Z. Harchaoui. Leveraging machine learning and accelerometry to classify animal behaviours with uncertainty. *Methods in Ecology and Evolution*, 17(2):567–584, 2026
- [3] **Medha Agarwal**, Z. Harchaoui, G. Mulcahy, and S. Pal. Langevin diffusion approximation to same marginal Schrödinger bridge. *Journal of Functional Analysis*, page 111494, 2026
- [4] **Medha Agarwal** and J. Xu. Quasi-Newton Acceleration of EM and MM Algorithms via Broyden’s Method. *Journal of Computational and Graphical Statistics*, 2023
- [5] **Medha Agarwal**, D. Vats, and V. Elvira. A principled stopping rule for importance sampling. *Electronic Journal of Statistics*, 2022
- [6] **Medha Agarwal** and D. Vats. Globally centered autocovariances in MCMC. *Journal of Computational and Graphical Statistics*, 2022

PREPRINTS

- [7] **Medha Agarwal**, Z. Harchaoui, G. Mulcahy, and S. Pal. Iterated Schrödinger bridge approximation to Wasserstein gradient flows. *arXiv preprint arXiv:2406.10823*, 2024

RESEARCH POSITIONS

JUN-DEC 2025	Abrahms Lab , Research Intern, PI: Briana Abrahms, <i>Seattle, WA</i>
MAR-JUN 2024	eScience Institute , Research Assistant, PI: Zaid Harchaoui & Briana Abrahms, <i>Seattle, WA</i>
2023	Institute for Foundations of Data Science , Research Assistant, PI: Zaid Harchaoui, <i>Seattle, WA</i>
JUL-SEPT 2020	University of Edinburgh , Research Intern, PI: Victor Elvira, <i>Edinburgh, Scotland</i>
MAY-JUL 2020	Duke University , Research Intern, PI: Jason Xu, <i>Durham, NC</i>

HONORS AND AWARDS

2025	Graduate Student Conference Presentation Award , University of Washington
2024	Winner, Student Paper Competition , ASA Conference on Statistical Learning and Data Science
2023	Hannan Graduate Student Award , Institute of Mathematical Statistics
2022	Center for Statistics and the Social Sciences Travel Award , University of Washington
2022	Scholarship recipient, Institute for Foundations of Data Science <i>Supported by the NSF Transdisciplinary Research in Principles of Data Science (TRIPODS) program</i>

2021	Department Fellowship, UW Statistics
2021	Proficiency Medal , Department of Mathematics and Statistics, IIT Kanpur <i>Awarded for the best academic performance among graduating students in each department</i>
2017	Academic Excellence Award - Dr. Sangeeta Goel Memorial Award , IIT Kanpur <i>Awarded for the highest womens' All India Rank in IIT-Joint Entrance Exam</i>

RESEARCH TALKS & PRESENTATIONS

★ Invited talk ● Oral presentation ◇ Poster

Sinkhorn Treatment Effects: A Causal Optimal Transport Measure

[1] NOV 2026	ASA Section on Statistical Learning and Data Science Conference ★	Brooklyn, NY
[2] JUL 2026	International Conference on Machine Learning (ICML) ◇	Seoul, South Korea
[3] MAY 2026	Optimal Transport + Optimization Workshop, NCCR SwissMAP ●	Les Diablerets, Switzerland

Iterated Schrödinger Bridge Approximation to Heat Flow

[4] AUG 2025	Joint Statistical Meetings ● <i>Advances in Generative Models Session</i>	Nashville, TN
[5] FEB 2025	Institute for Foundations of Machine Learning ● <i>Mathematics of Deep Learning Workshop</i>	Austin, TX
[6] JAN 2025	Joint Mathematics Meetings ★ <i>Mathematics of Adversarial, Interpretable, and Explainable AI</i>	Seattle, WA
[7] NOV 2024	ASA Conference on Statistical Learning and Data Science ◇	Newport Beach, CA
[8] OCT 2024	SIAM Mathematics of Data Science ◇	Atlanta, GA

Leveraging Machine Learning and Accelerometry to Classify Animal Behaviors

[9] OCT 2025	Institute for Foundations of Data Science Seminar ●	Seattle, WA
[10] FEB 2025	UW Data Science Seminar ★ <i>Seminar Series for AI@UW Seed Grant Awardees</i>	Seattle, WA

Globally centered autocovariances in MCMC

[11] MAY 2023	BayesComp ★ <i>Session on Measuring Quality of MCMC Samples</i>	Levi, Finland
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SELECTED WORKSHOPS ATTENDED

† Invited attendee

JUL 2025	Wasserstein Gradient Flows in Math and ML Workshop † <i>Banff International Research Station</i>	Banff, Canada
AUG 2022	Deep Learning Theory Workshop and Summer School <i>Simons Institute for the Theory of Computing</i>	Berkeley, CA
JUL 2020	noRth 2020: A Virtual Conference for R Users	Minneapolis, MN

TEACHING EXPERIENCE

SPRING 2026	STAT 498/591 - Special Topics: Optimization and Sampling
WINTER 2026	STAT 311 - Elements of Statistical Methods
SPRING 2025	CSE 546 - Machine Learning for Big Data
AUTUMN 2024	STAT 512 - Statistical Inference I
WINTER 2024, 2023	STAT 513 - Statistical Inference II
2021–2022	STAT 311 - Elements of Statistical Methods

INDUSTRY EXPERIENCE

SUMMER 2026	Quantbot Technologies QUANTITATIVE RESEARCHER INTERN ● Developing statistical learning methods for equity return prediction, with an emphasis on modeling idiosyncratic stock movements.	New York City, NY
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SUMMER 2024	Amazon Science APPLIED SCIENTIST INTERN Developed multi-horizon probabilistic forecasting methods for multivariate time series using MQ-CNN; achieved > 95% empirical coverage on 5+ years of production-scale demand data.	<i>New York City, NY</i>
SUMMER 2023	Amazon Science APPLIED SCIENTIST INTERN - Fine-tuned a 40B foundation model using reinforcement learning from human feedback, integrating human preferences with product-description corpora. - Project paper featured in Amazon Machine Learning Conference 2023.	<i>Bellevue, WA</i>

PROFESSIONAL SERVICE

Reviewer, *Journal of Statistical Planning and Inference*, 2026

ACADEMIC SERVICE

2022-23	Leader, Normalizing Flows Working Group (Webpage)	<i>UW Seattle</i>
WINTER 2021	Statistics Undergraduate Directed Reading Program (SPA-DRP) Mentor	<i>UW Seattle</i>
AUTUMN 2021	Pre-application Review Service (PARS) volunteer	<i>UW Seattle</i>
2019-20	Student Undergraduate Committee Student Nominee	<i>IIT Kanpur</i>